

# Astro-3\_M\_down CC



# Astro-3\_M\_down BP

myelin assembly

ensheathment of neurons

myelin assembly

axon ensheathment

myelination

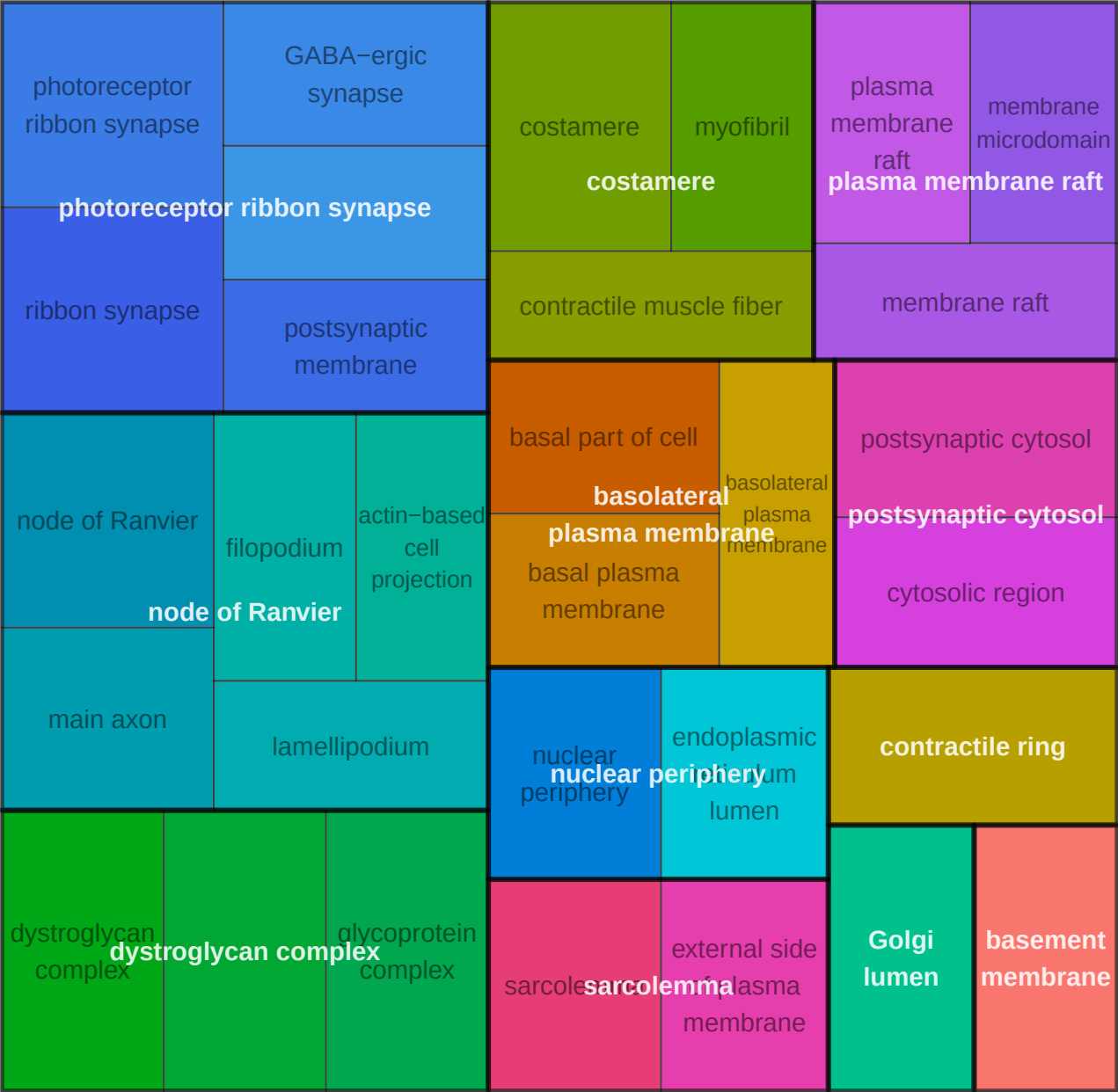
positive  
regulation  
of bone  
resorption

regulation of endocytosis

regulation of endocytosis

regulation of vesicle-mediated transport

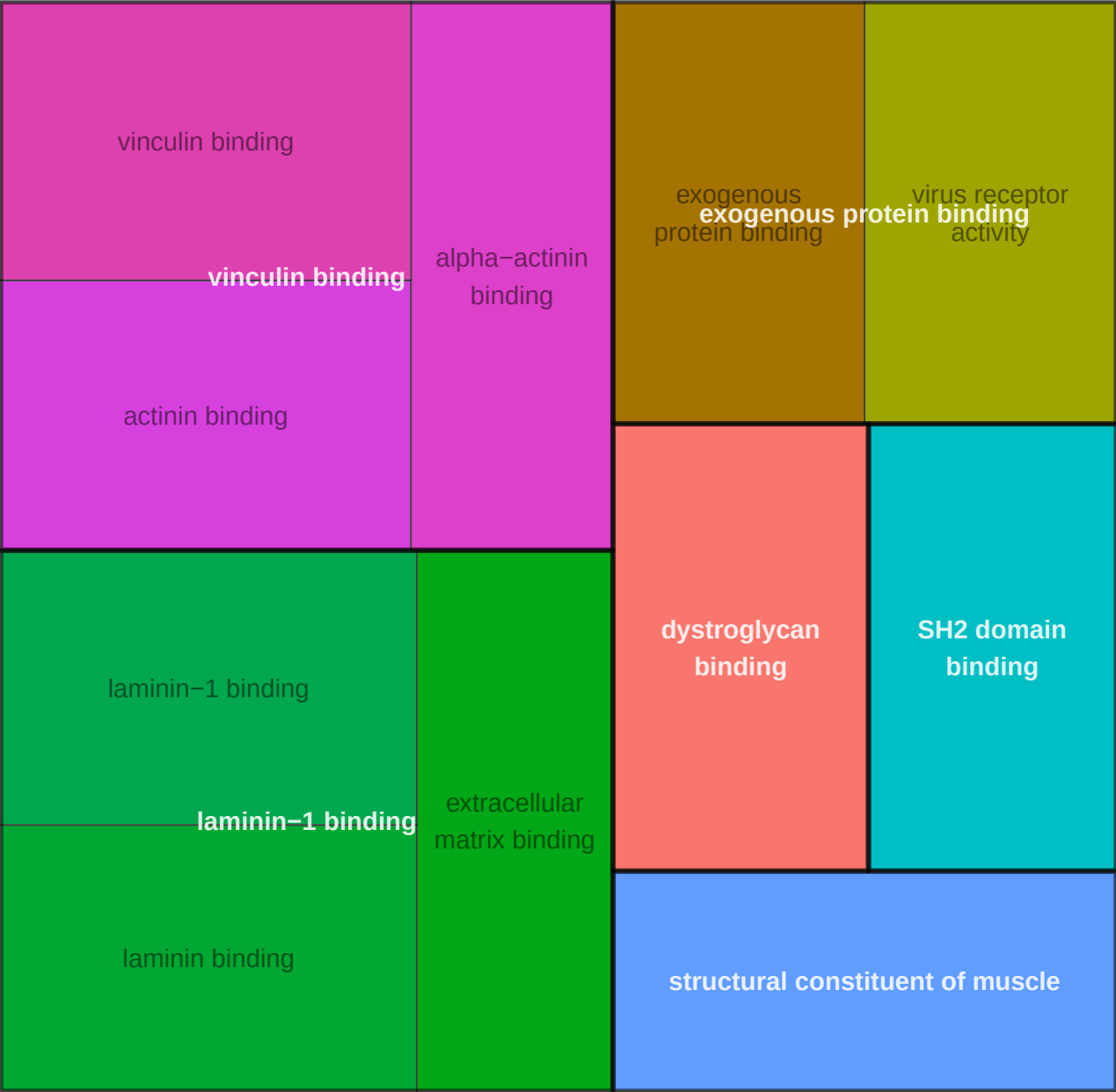
Astro-3\_M\_up CC



# Astro-3\_M\_up BP



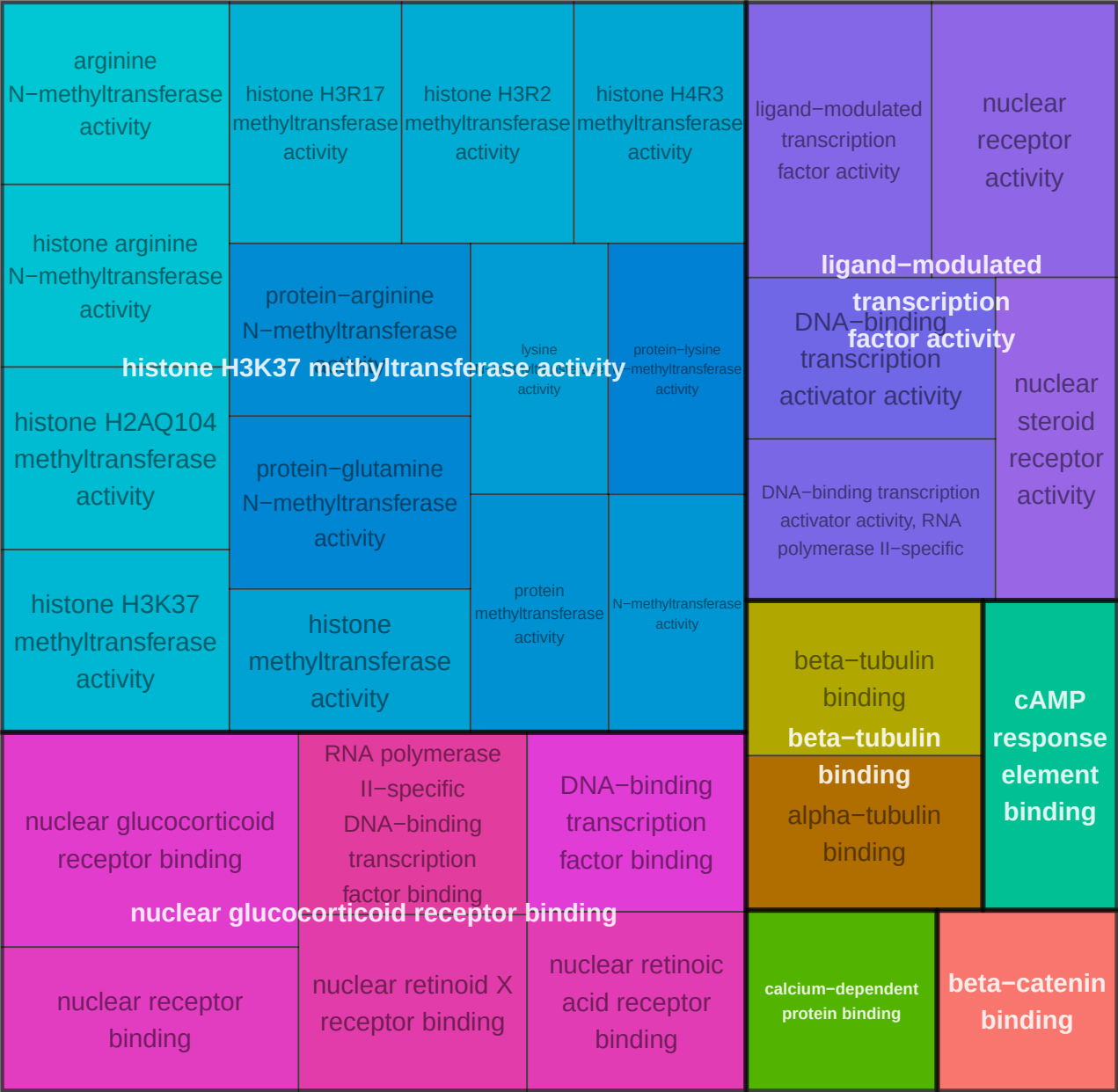
Astro-3\_M\_up MF



Astro-4\_F\_up BP



Astro-4\_F\_up MF



Excit-L2\_5-1\_0\_downF\_DownM BP

|                                                    |                                                                |                                                              |                                                       |                                                       |                                                                      |
|----------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------|
| endosome to plasma membrane protein transport      | establishment of protein localization to postsynaptic membrane | neurotransmitter receptor transport to postsynaptic membrane | positive regulation of neuron projection arborization | neuron projection arborization                        | dendritic spine development                                          |
| protein localization to postsynapse                | endosome to plasma membrane protein transport                  | establishment of protein localization to plasma membrane     | regulation of neuron projection arborization          | positive regulation of neuron projection arborization | dendritic spine organization                                         |
|                                                    |                                                                |                                                              |                                                       |                                                       | neuron projection organization                                       |
| protein localization to postsynaptic membrane      | protein localization to synapse                                | protein localization to cell junction                        | acetyl-CoA biosynthetic process                       | acetyl-CoA metabolic process                          | neurotransmitter receptor transport to plasma membrane               |
| positive regulation of peptidyl-lysine acetylation | positive regulation of peptidyl-lysine acetylation             | regulation of protein acetylation                            | acyl-CoA biosynthetic process                         | thioester biosynthetic process                        | neurotransmitter receptor transport, endosome to plasma membrane     |
|                                                    |                                                                |                                                              |                                                       |                                                       | neurotransmitter receptor transport, endosome to plasma membrane     |
|                                                    |                                                                |                                                              |                                                       |                                                       | receptor localization to synapse                                     |
| positive regulation of peptidyl-lysine acetylation | positive regulation of peptidyl-lysine acetylation             | protein acetylation                                          | acyl-CoA metabolic process                            | thioester metabolic process                           | regulation of postsynaptic membrane neurotransmitter receptor levels |
| regulation of peptidyl-lysine acetylation          | protein acetylation                                            | protein acylation                                            | neuropeptide signaling pathway                        |                                                       |                                                                      |



# Excit-L2\_5-1\_0\_downF\_DownM MF

acid-thiol ligase activity

ligase activity, forming  
carbon-sulfur bonds

CoA-ligase activity

CoA-ligase activity

ligase activity

beta-catenin  
binding

signaling receptor  
complex adaptor activity

signaling receptor complex adaptor activity

# Excit-L2\_5-1\_0\_upF\_DownM CC

GINS complex

GINS complex

DNA replication preinitiation complex

protein-DNA  
complex

nuclear chromosome

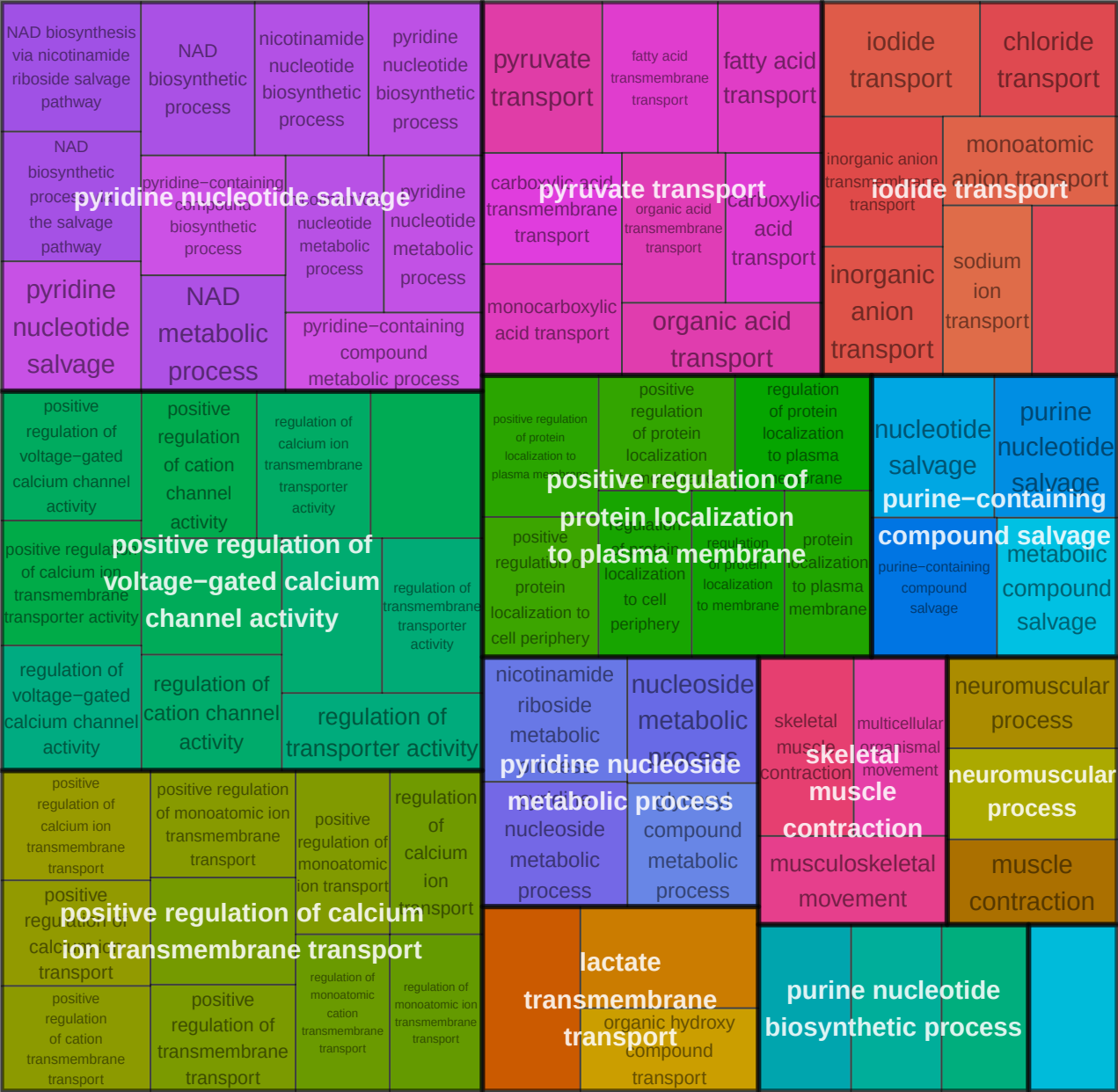
sarcolemma

cytoplasmic side of  
plasma membrane

cytoplasmic side of  
plasma membrane

cytoplasmic side of  
plasma membrane

Excit-L2\_5-1\_0\_upF\_DownM BP



Excit-L2\_5-1\_0\_upF\_DownM MF

|                                               |                                                                     |                                                     |                                                  |                                                     |                                                        |
|-----------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|
| iodide transmembrane transporter activity     | secondary active monocarboxylate transmembrane transporter activity |                                                     | organic acid:sodium symporter activity           |                                                     | monocarboxylic acid transmembrane transporter activity |
|                                               | lactate transmembrane transporter activity                          | solute:sodium symporter activity                    | monoatomic anion channel activity                | solute:monoatomic cation symporter activity         | carboxylic acid transmembrane transporter activity     |
| iodide transmembrane transporter activity     |                                                                     |                                                     |                                                  |                                                     |                                                        |
| fatty acid transmembrane transporter activity | inorganic anion transmembrane transporter activity                  | monoatomic anion transmembrane transporter activity | sodium ion transmembrane transporter activity    | symporter activity                                  |                                                        |
| monocarboxylate:sodium symporter activity     | lipid transporter activity                                          | organic acid transmembrane transporter activity     | organic anion transmembrane transporter activity | secondary active transmembrane transporter activity |                                                        |

## Excit-L2\_5-1\_M\_up CC

specific granule membrane

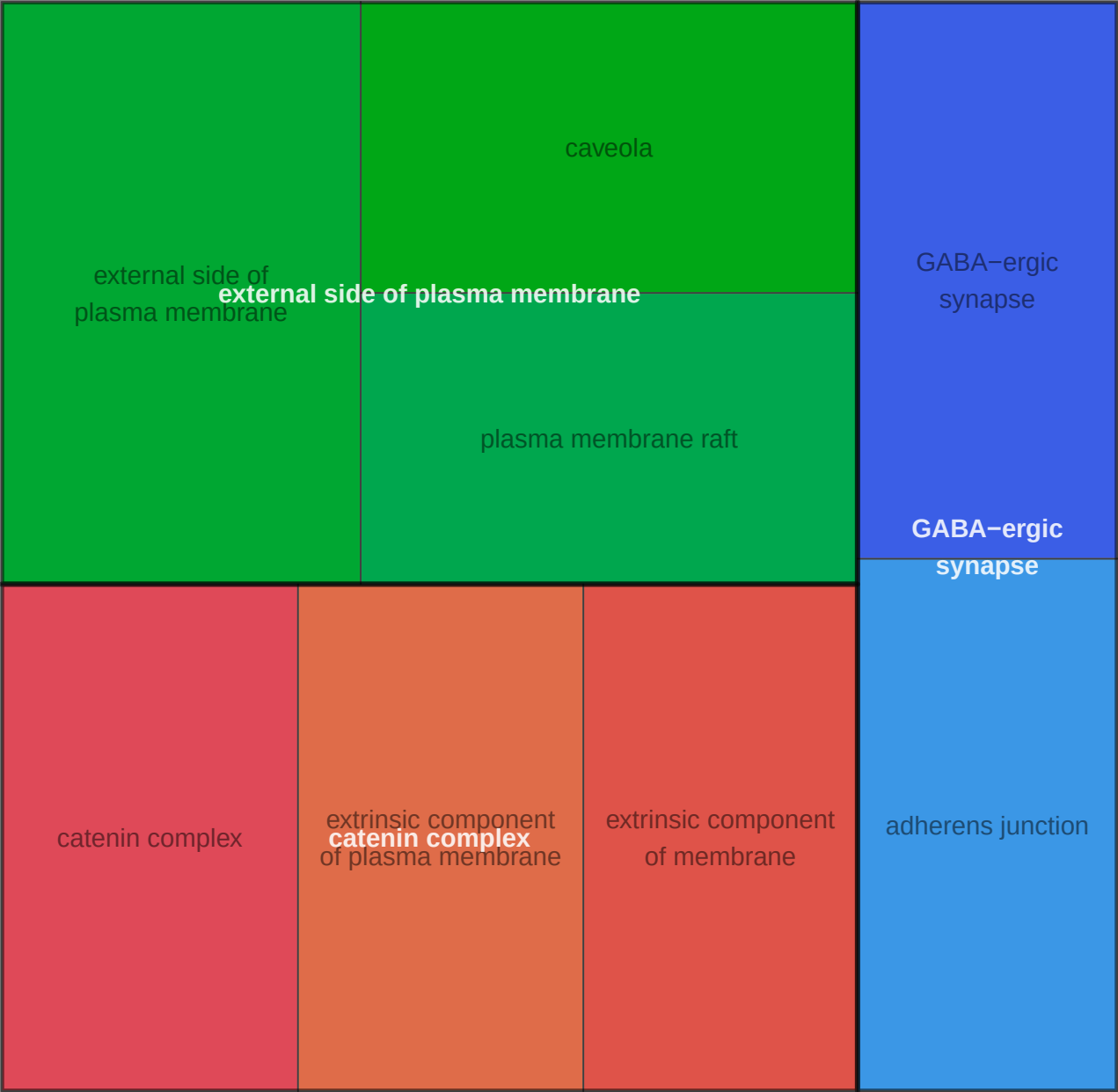
**specific granule membrane**

secondary granule membrane

# Excit-L2\_5-1\_M\_up MF



Excit-L2\_5-2\_M\_down CC



Excit-L2\_5-2\_M\_down BP





Excit-L5-1\_0\_downF\_UpM BP



# Excit-L5-1\_0\_downF\_UpM MF

hormone binding

**hormone binding**

peptide hormone binding

amide binding

G protein-coupled peptide  
receptor activity

**peptide receptor activity**

peptide receptor activity

# Excit-L5-2\_0\_downF\_UpM CC



Excit-L5-2\_0\_downF\_UpM BP



Excit-L5-2\_0\_downF\_UpM MF



# Excit-L5-2\_M\_down CC

germ cell nucleus

germ cell nucleus

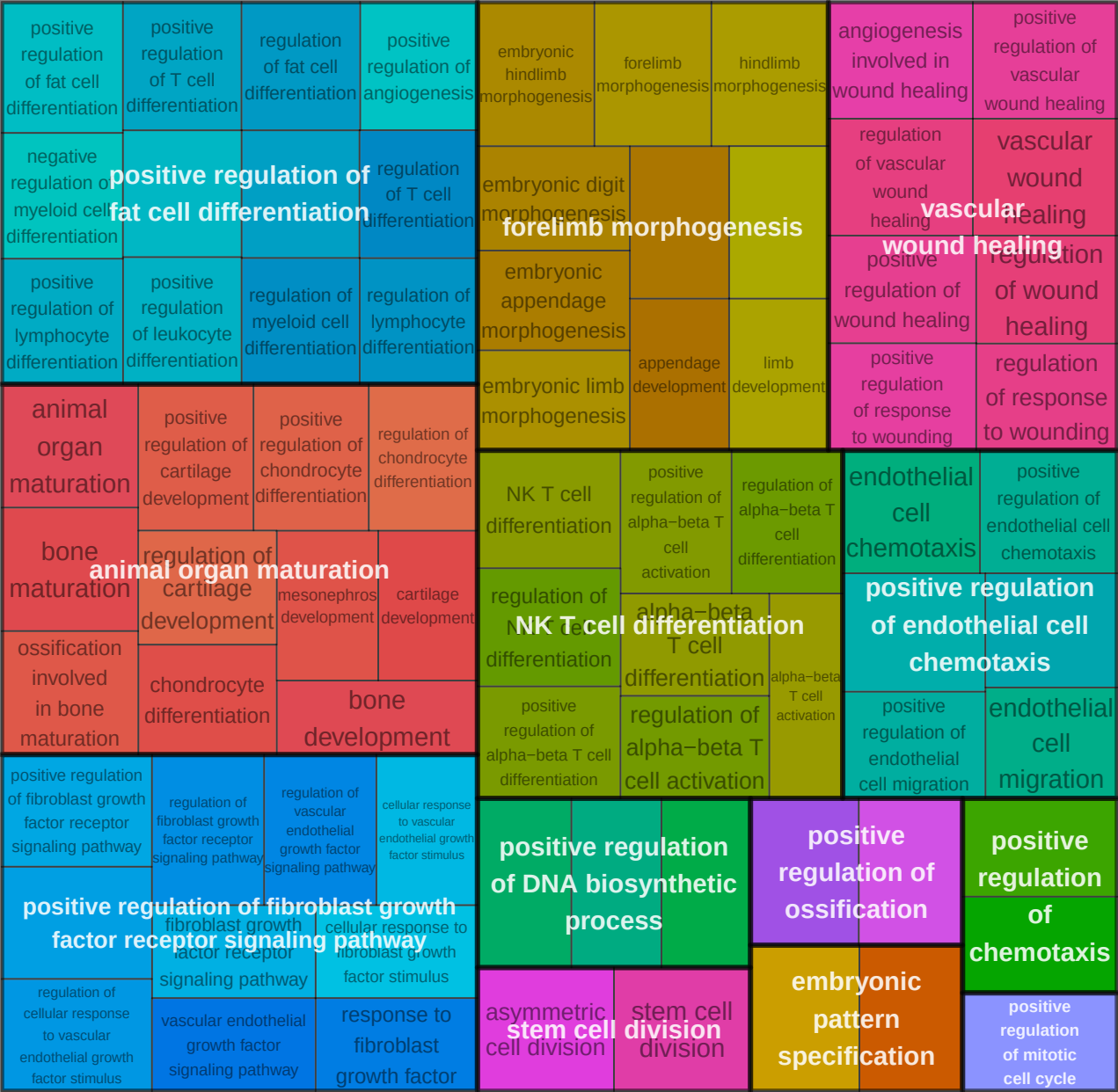
male germ cell nucleus

PML body

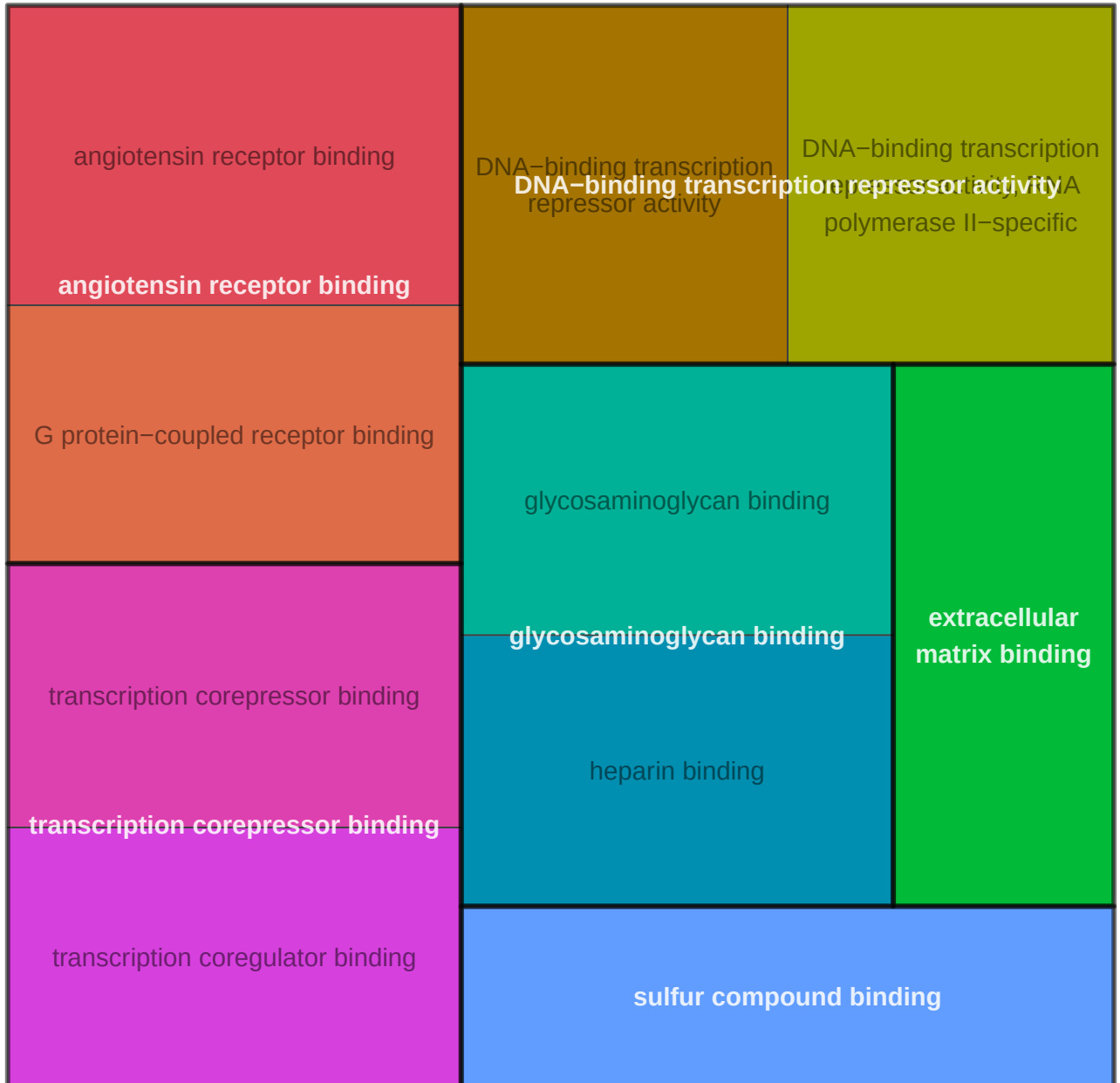
transcription repressor complex

basement membrane

Excit-L5-2\_M\_down BP



# Excit-L5-2\_M\_down MF





Excit-L5-2\_M\_up CC

adherens junction

**adherens junction**

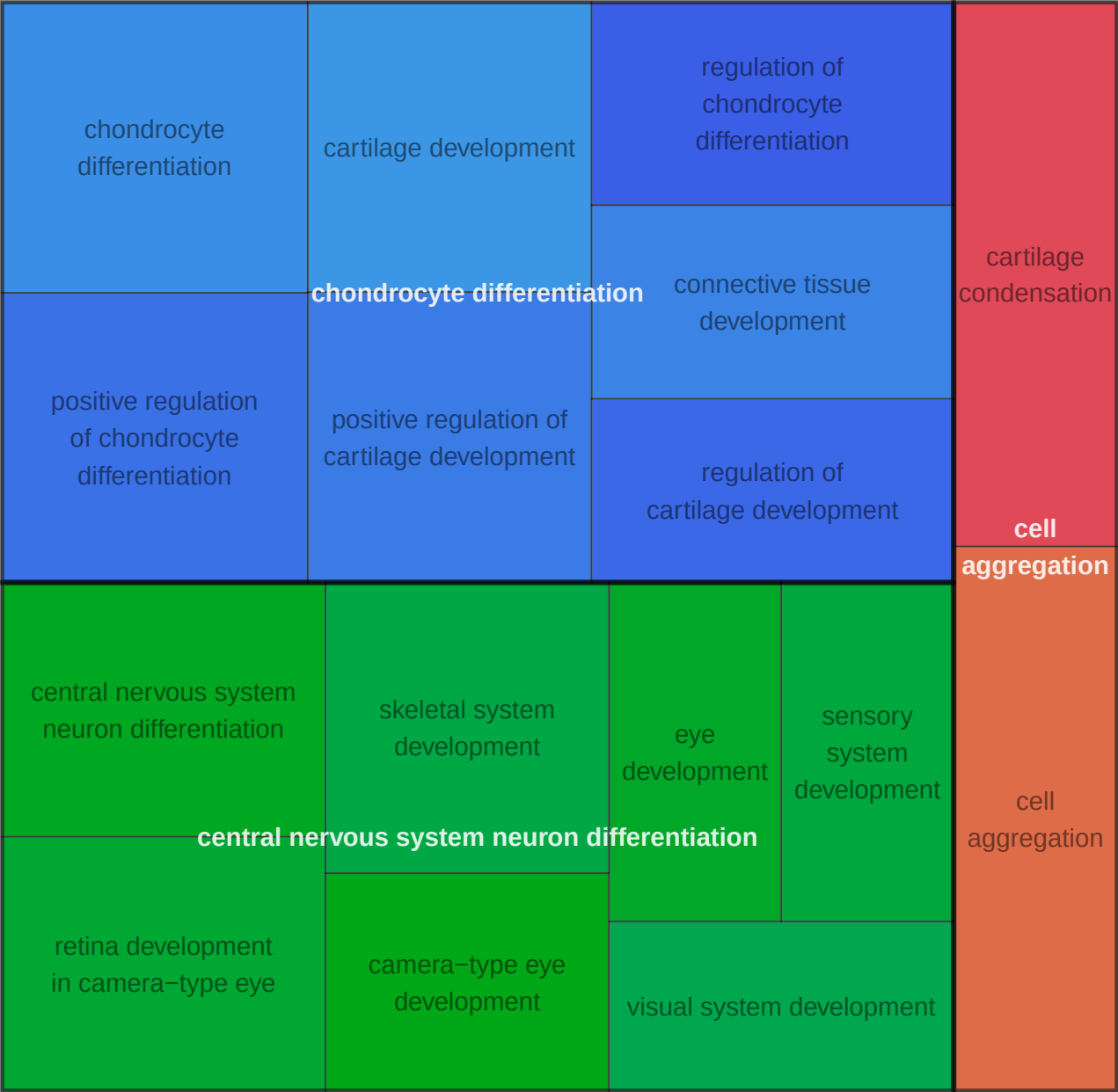
apical junction complex

cell-cell junction

**HFE-transferrin  
receptor  
complex**

**cerebellar mossy fiber**

Excit-L5-2\_M\_up BP



# Macro\_M\_down CC

cell-substrate junction

semaphorin receptor complex

cell-substrate junction

focal adhesion

sorting endosome

neurofilament

cell body membrane

# Macro\_M\_down BP



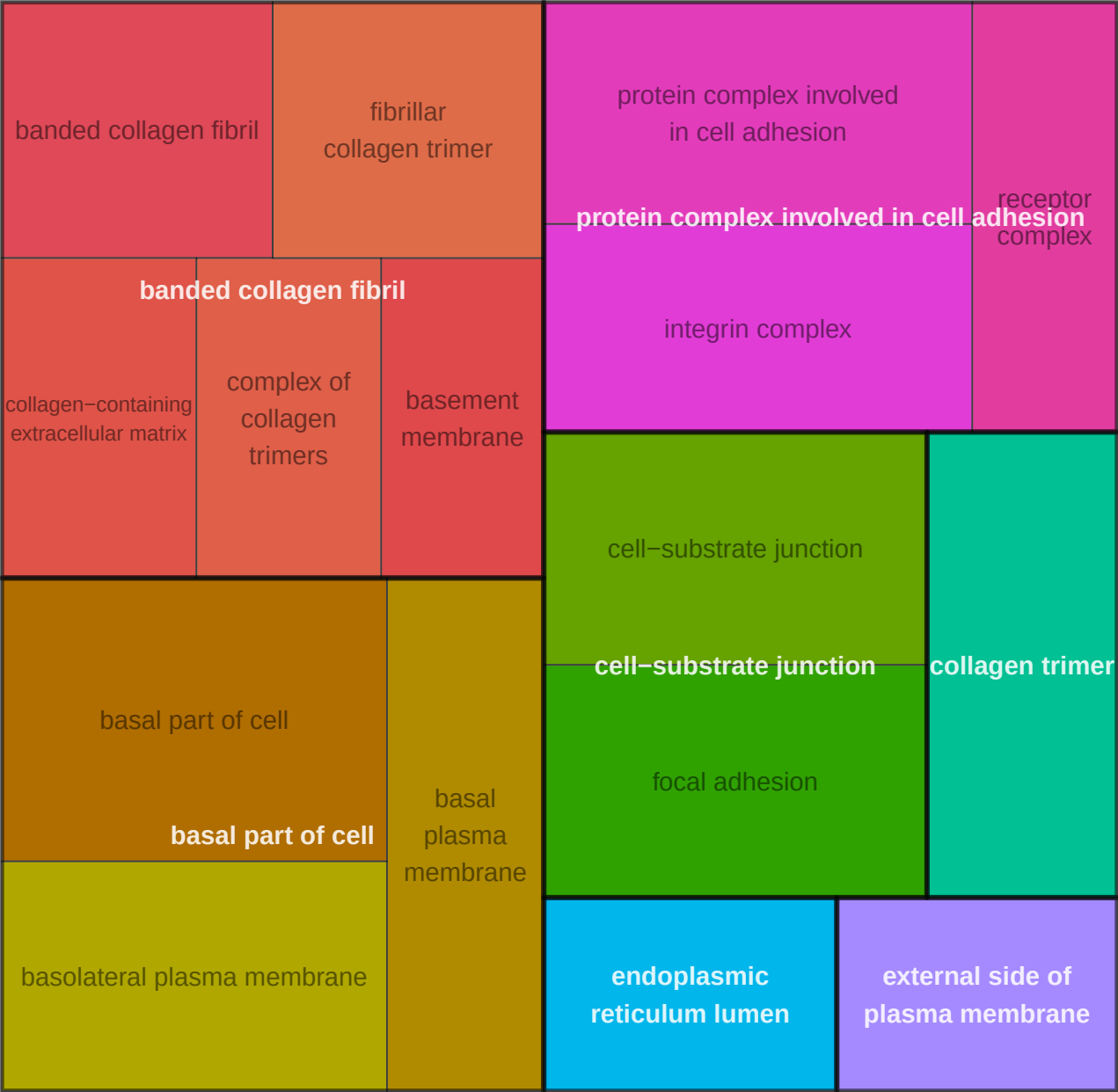
Macro\_M\_down MF



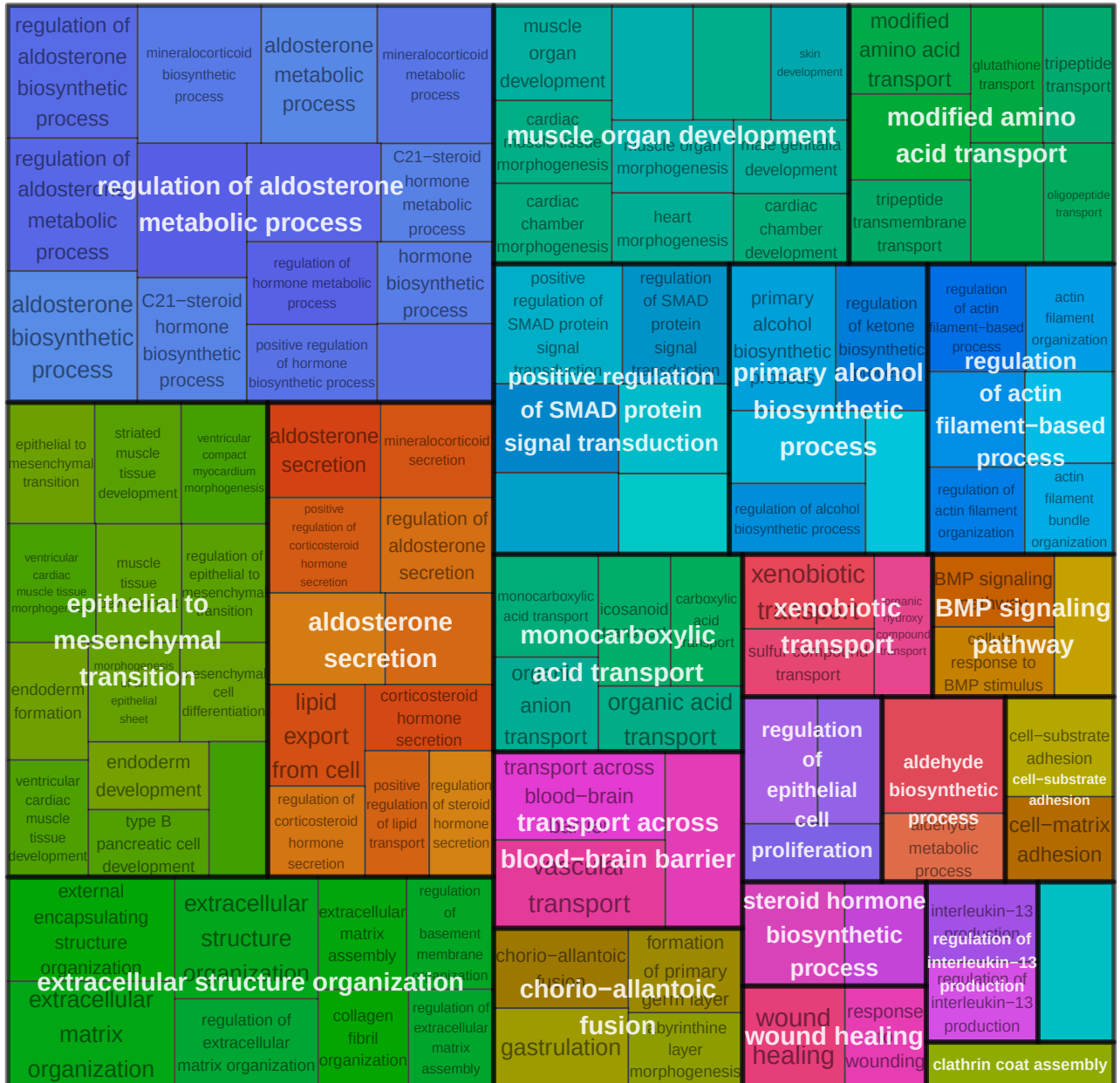
# Macro\_M\_up MF



Oligo-3\_0\_downF\_UpM CC

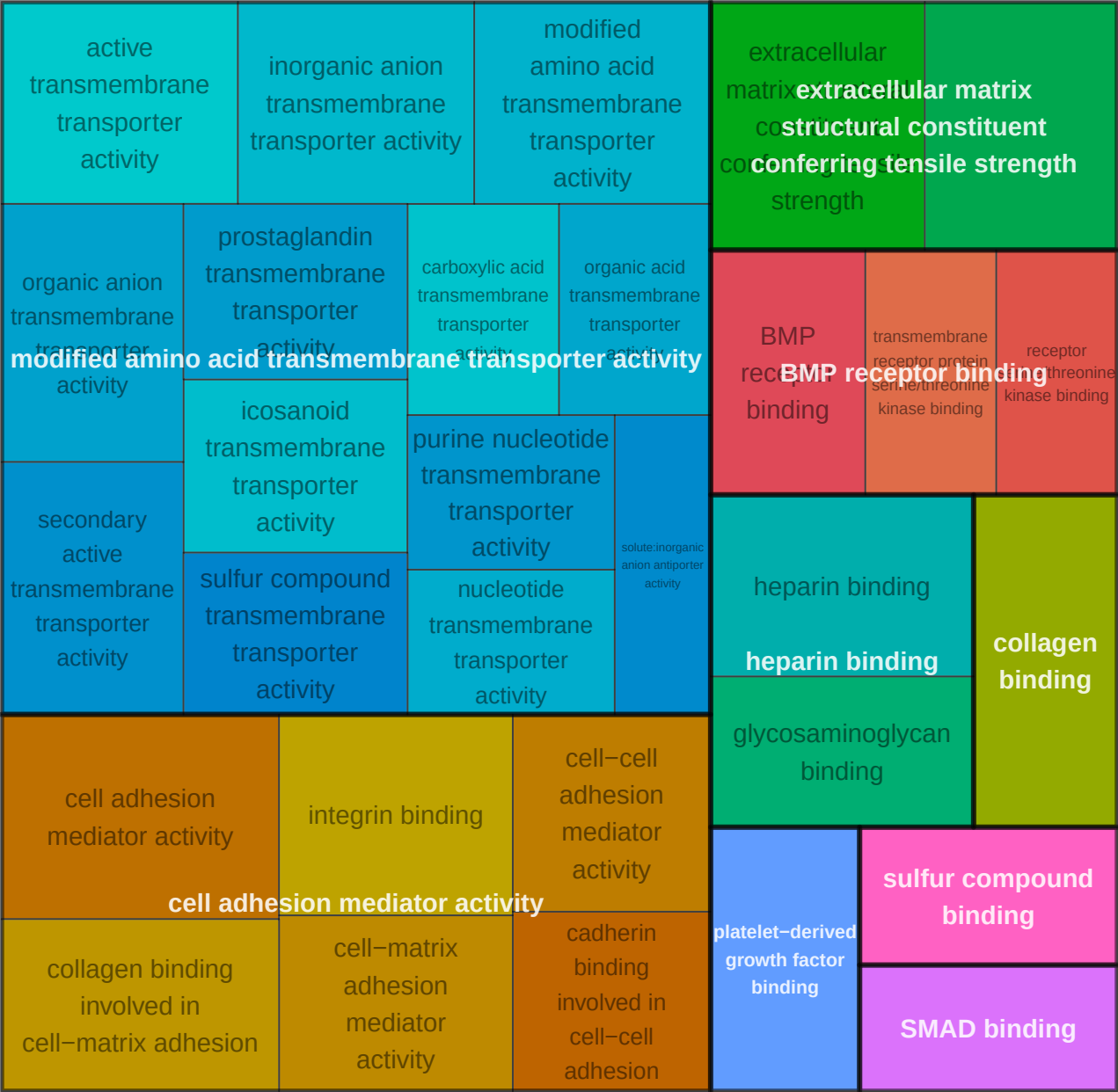


Oligo-3\_0\_downF\_UpM BP





Oligo-3\_0\_downF\_UpM MF



Oligo-3\_0\_upF\_DownM CC

collagen type IX trimer **FACIT collagen trimer** FACIT collagen trimer

Oligo-3\_0\_upF\_DownM BP

deoxyribonucleoside  
triphosphate  
catabolic process

purine nucleoside  
triphosphate  
catabolic process

deadenylation-independent  
decapping of  
nuclear-transcribed mRNA

purine nucleoside triphosphate catabolic process

**nuclear-transcribed  
mRNA catabolic  
process,  
deadenylation-independent  
decay**

deoxyribonucleoside  
triphosphate  
metabolic process

nucleoside triphosphate  
catabolic process

nuclear-transcribed  
mRNA catabolic  
process,  
deadenylation-independent  
decay

# Oligo-3\_0\_upF\_DownM MF

cis-trans isomerase activity

**cis-trans isomerase activity**

peptidyl-prolyl cis-trans isomerase activity

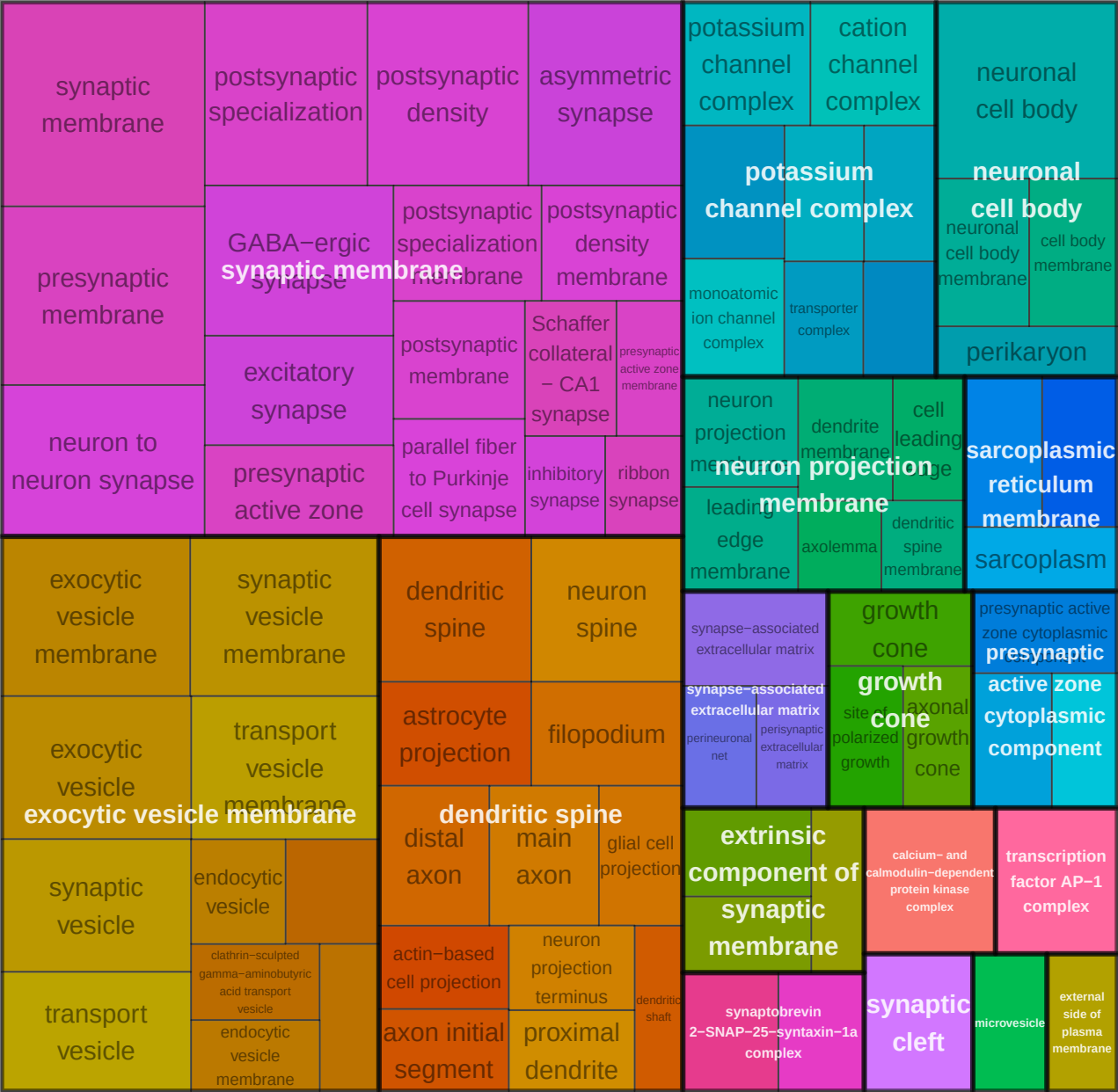
**cyclosporin A binding**

**nucleoside triphosphate  
diphosphatase activity**

**extracellular matrix structural  
constituent conferring tensile strength**

**nucleosome binding**

Oligo-3\_0\_upF\_UpM CC



Oligo-3\_0\_upF\_UpM BP



## Oligo-3\_0\_upF\_UpM MF

voltage-gated monoatomic ion  
channel activity involved in  
regulation of presynaptic  
membrane potential

**voltage-gated monoatomic ion channel activity involved  
in regulation of presynaptic membrane potential**

metal ion transmembrane  
transporter activity

**glutamate receptor binding**

monoatomic cation channel activity

**calcium-dependent phospholipid binding**

**SNARE binding**

# Oligo-3\_M\_down CC

myelin sheath **myelin sheath**

compact myelin

neuromuscular junction

**neuromuscular junction**

cell-cell junction

main axon **main axon**

node of Ranvier

**cell cortex**



## Oligo-3\_M\_down BP

axon ensheathment

myelination

axon  
ensheathment  
in central  
nervous system

axon ensheathment

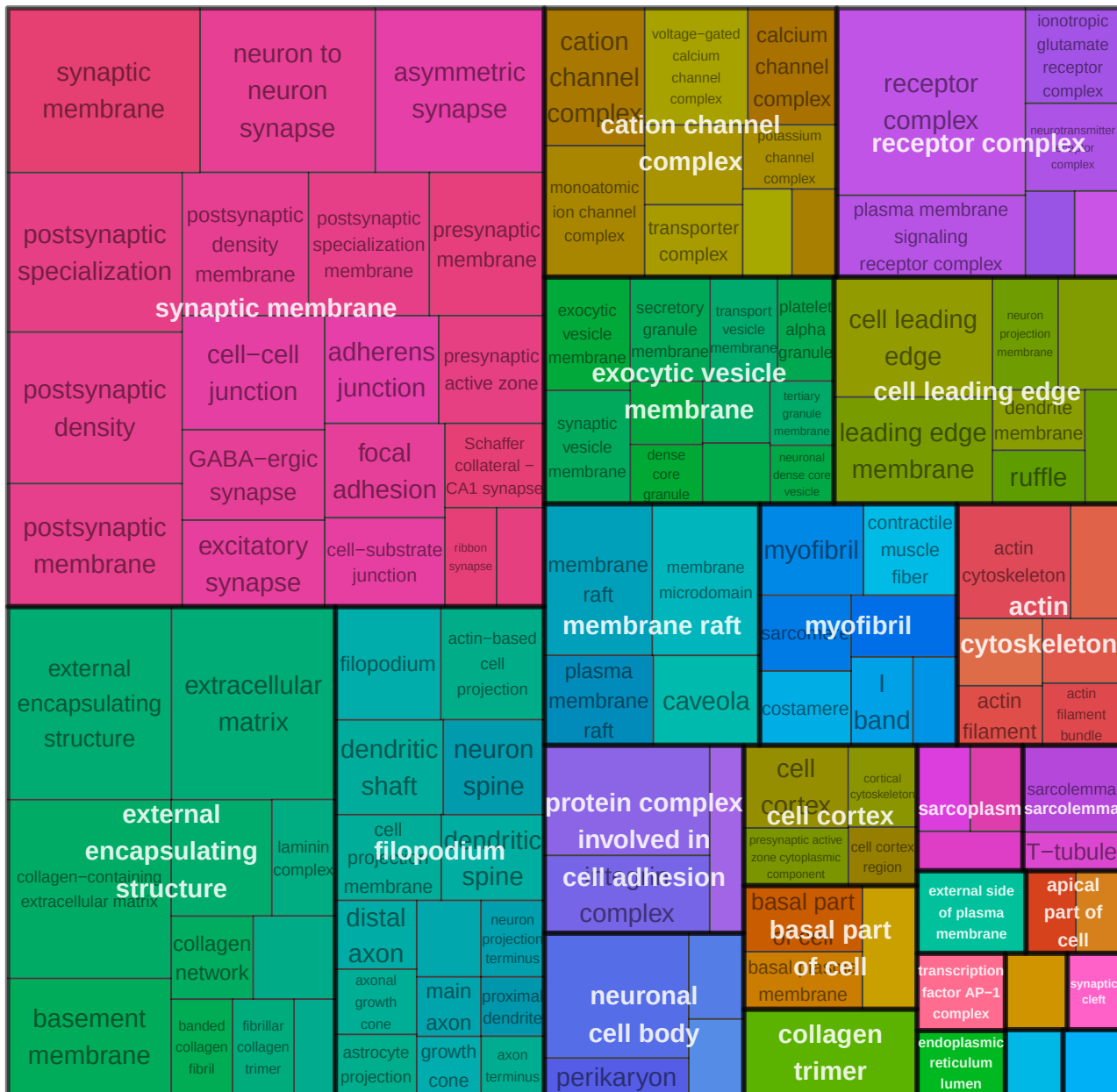
central nervous  
system myelination

ensheathment of neurons

oligodendrocyte  
development

myelin assembly

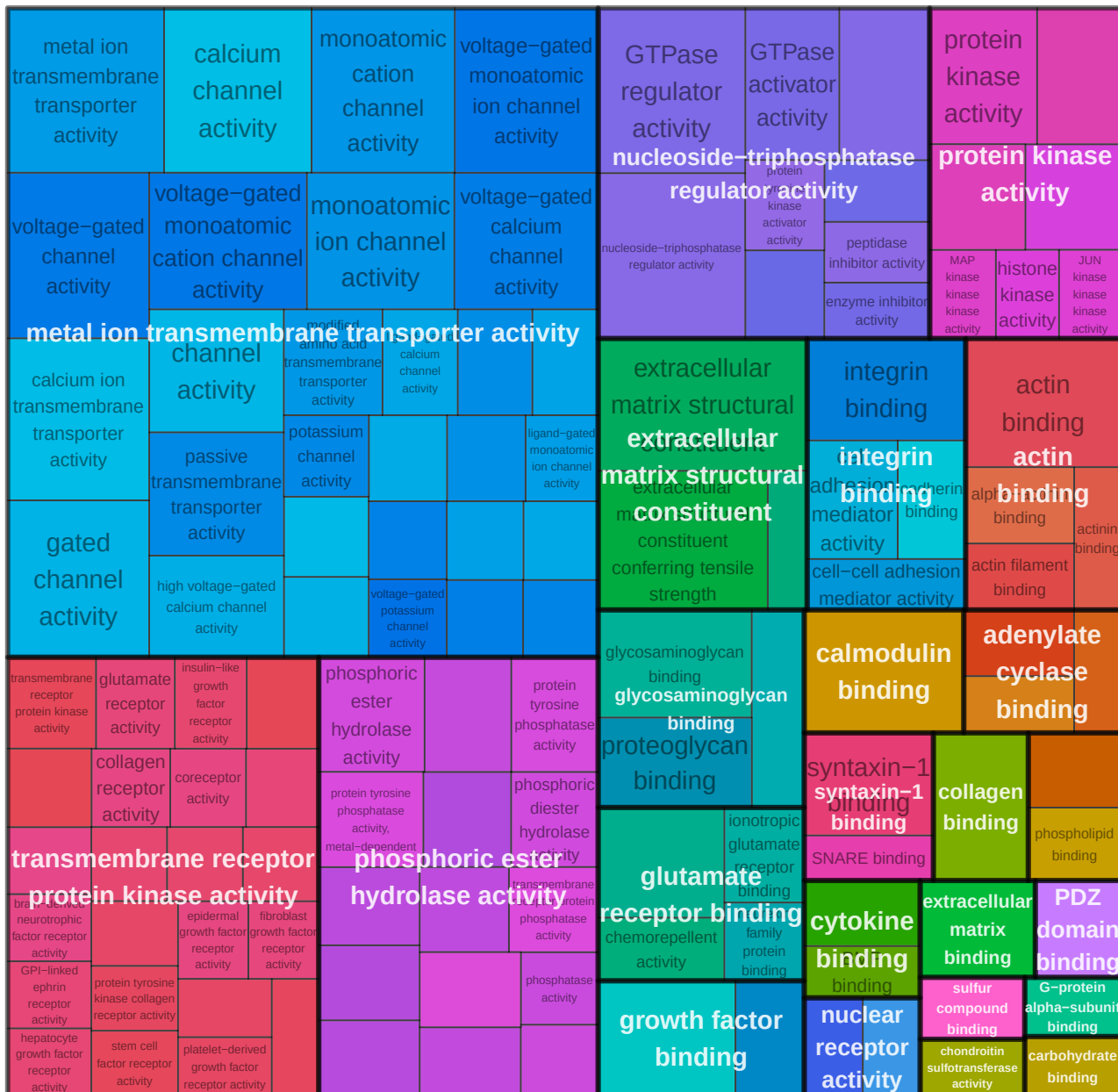
Oligo-3 M up CC



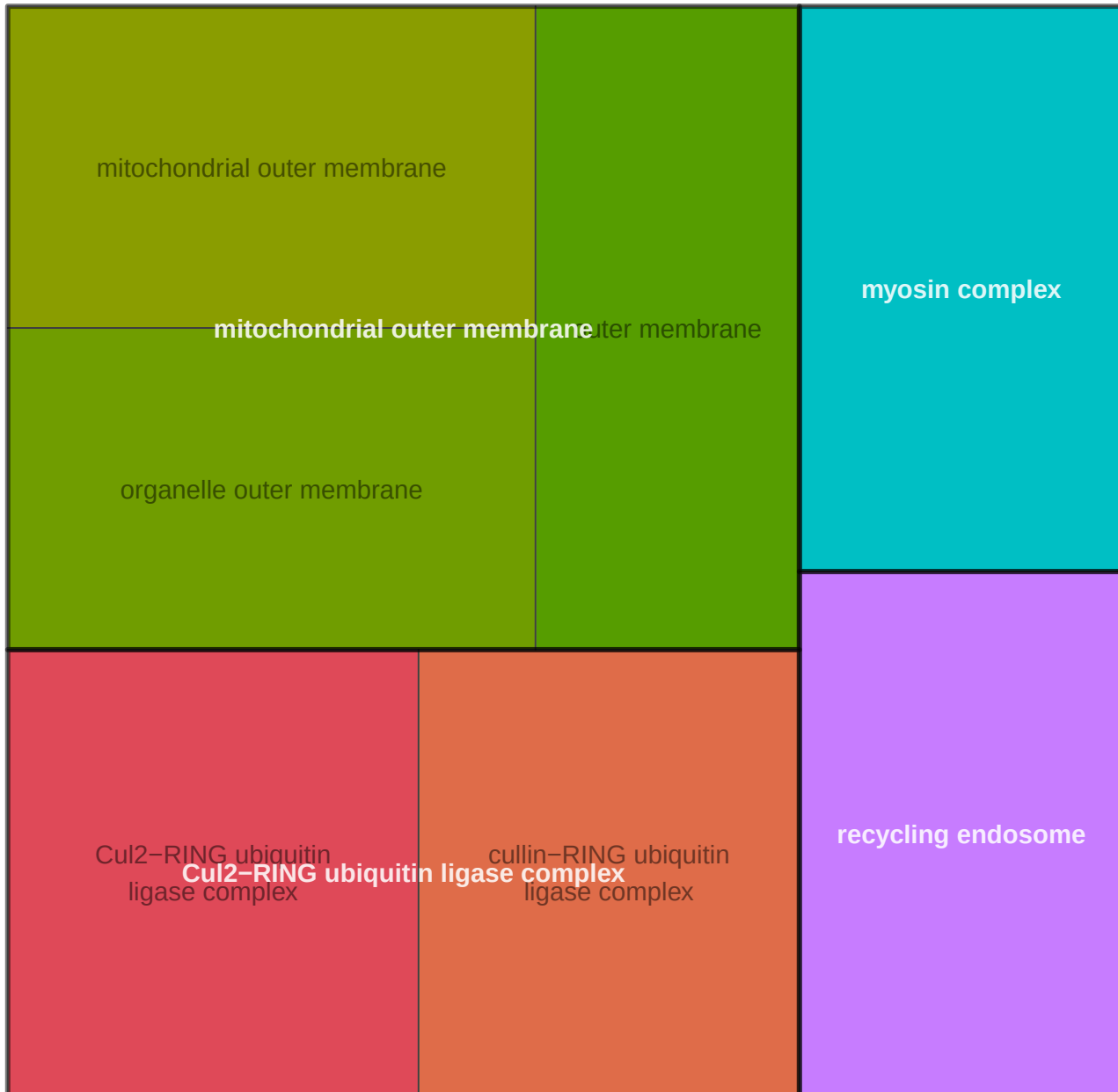
# Oligo-3\_M\_up BP



# Oligo-3\_M\_up MF



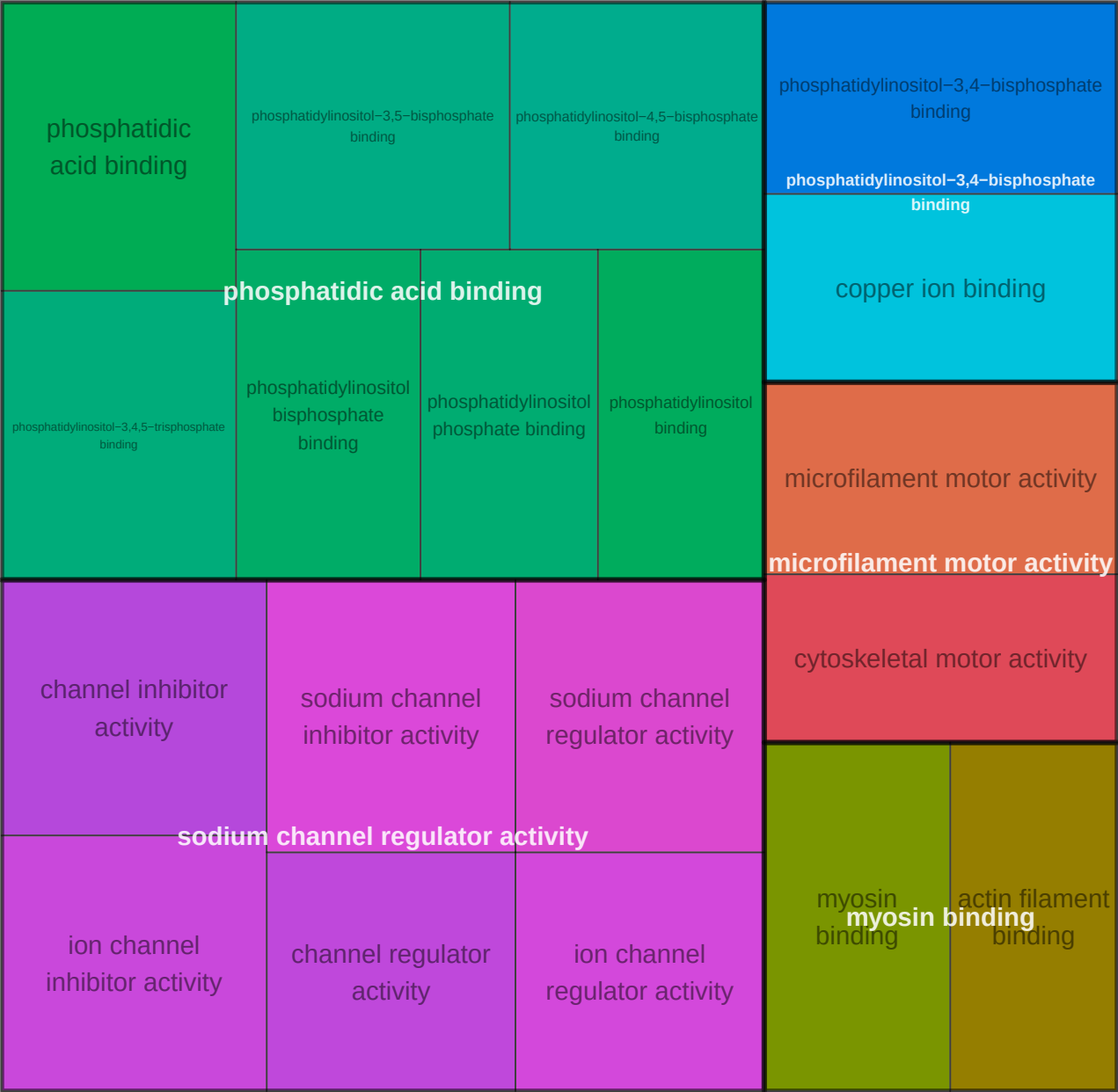
# Oligo-5\_0\_downF\_DownM CC



# Oligo-5\_0\_downF\_DownM BP



Oligo-5\_0\_downF\_DownM MF



Oligo-5\_0\_upF\_UpM MF

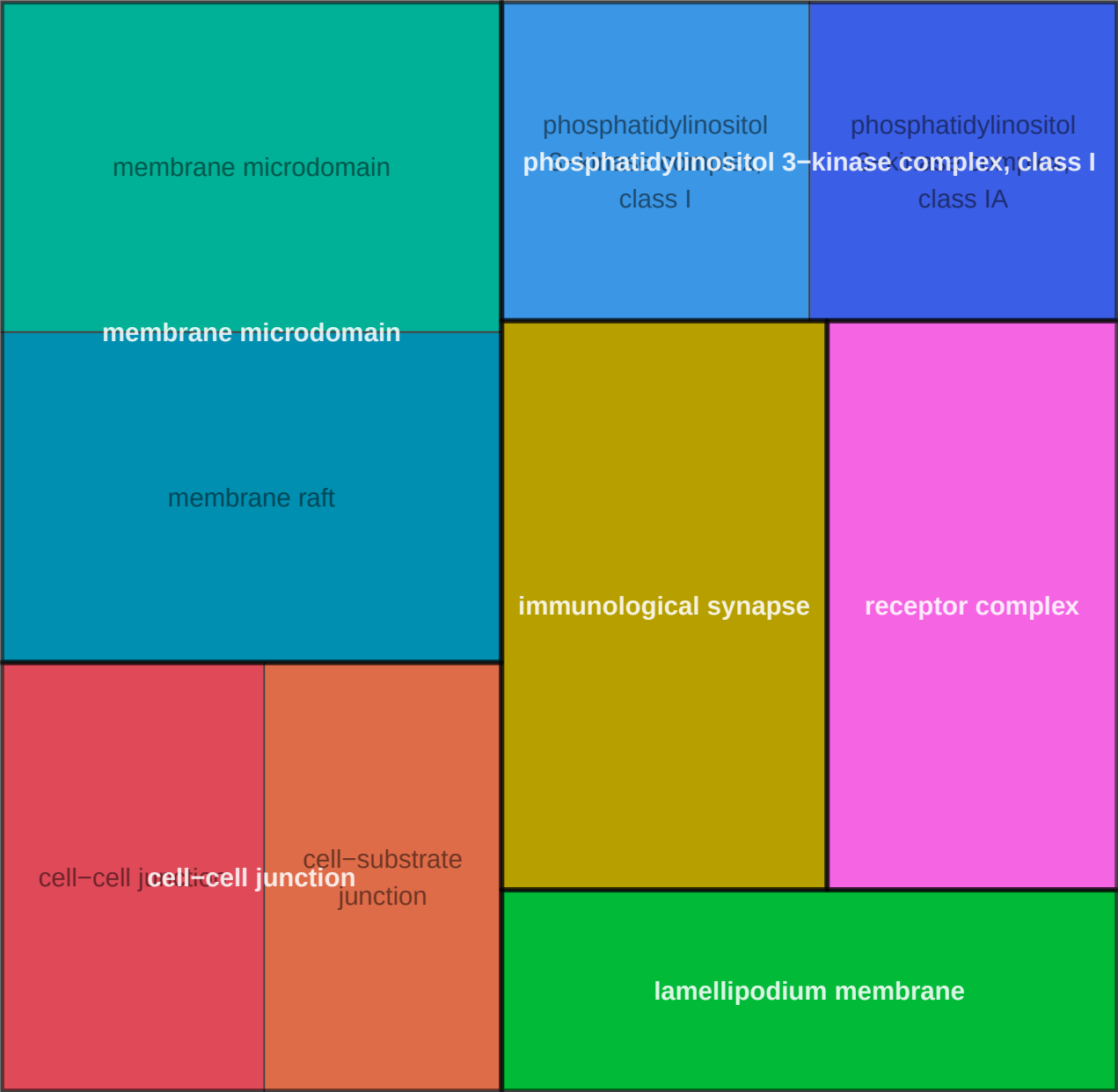
|                                                     |                                              |                                                        |                                                      |                                                         |                                                    |
|-----------------------------------------------------|----------------------------------------------|--------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------|
| adrenergic receptor activity                        | ephrin receptor activity                     | epidermal growth factor receptor activity              | fibroblast growth factor receptor activity           | G protein-coupled amine receptor activity               | neurotrophin binding                               |
| alpha-adrenergic receptor activity                  | GPI-linked ephrin receptor activity          | macrophage colony-stimulating factor receptor activity | neurotrophin receptor activity                       | placental growth factor receptor activity               |                                                    |
| boss receptor activity                              | hepatocyte growth factor receptor activity   | platelet-derived growth factor activity                | protein tyrosine kinase collagen receptor activity   |                                                         | S100 protein binding                               |
| brain-derived neurotrophic factor receptor activity | insulin receptor activity                    | platelet-derived growth factor beta-receptor activity  | transmembrane-ephrin receptor activity               | transmembrane receptor protein tyrosine kinase activity |                                                    |
| collagen receptor activity                          |                                              |                                                        |                                                      | protein-hormone receptor activity                       |                                                    |
|                                                     | insulin-like growth factor receptor activity | platelet-derived growth factor receptor activity       | vascular endothelial growth factor receptor activity | transmembrane receptor protein kinase activity          | structural molecule activity conferring elasticity |
| membrane scission                                   |                                              | membrane scission GTPase motor activity                |                                                      | molecular sequestering activity                         |                                                    |



## Oligo-5\_M\_down CC

collagen type IX trimer collagen type IX trimer FACIT collagen trimer

Oligo-5\_M\_up CC





## Oligo-5\_M\_up MF

protein kinase activity

transcription coactivator binding

transmembrane receptor protein  
protein kinase activity  
tyrosine kinase activity

transcription coactivator binding

transcription coregulator binding

vascular endothelial growth factor receptor activity

## OPC-5\_F\_up CC

anaphase-promoting complex

anaphase-promoting complex

nuclear ubiquitin ligase complex

cullin-RING ubiquitin  
ligase complex

OPC-5\_F\_up BP



Vasc-Endo\_0\_downF\_DownM BP



# Vasc-Endo\_0\_downF\_DownM MF

calcium, potassium:sodium  
antiporter activity

calcium:monoatomic cation  
antiporter activity

calcium:monoatomic cation antiporter activity

STAT family  
protein binding

calcium:sodium antiporter activity

solute:potassium antiporter activity



# Vasc-Endo\_0\_downF\_UpM CC



|                                                                            |                                          |                                                                                                 |                                        |                                                                                        |                                                                                                  |                                     |                                                        |                                           |
|----------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------|-------------------------------------------|
| regulation of relaxation of cardiac muscle                                 | regulation of relaxation of muscle       | negative regulation of blood circulation                                                        | interleukin-5 production               | positive regulation of interleukin-5 production                                        | cAMP catabolic process                                                                           | cyclic nucleotide catabolic process | vestibulocochlear nerve development                    | cranial nerve morphogenesis               |
| negative regulation of heart contraction                                   | relaxation of cardiac muscle             | positive regulation of heart contraction                                                        | regulation of interleukin-5 production | positive regulation of interleukin-2 production                                        | cAMP catabolic process                                                                           |                                     | vestibulocochlear nerve development                    |                                           |
| positive regulation of heart rate                                          | relaxation of muscle                     | cell communication involved in cardiac conduction                                               | interleukin-2 production               | regulation of interleukin-2 production                                                 | cyclic nucleotide metabolic process                                                              | bile acid metabolic process         | inner ear receptor cell differentiation                | mechanoreceptor differentiation           |
| calcium ion transmembrane transport via high voltage-gated calcium channel | regulation of monoatomic anion transport | regulation of monoatomic ion transport                                                          | aminophospholipid translocation        | apical protein localization                                                            | negative regulation of adenylate cyclase-activating G protein-coupled receptor signaling pathway |                                     | regulation of plasma membrane organization             | phospholipid translocation                |
|                                                                            |                                          | release of sequestered calcium ion into cytosol by sarcoplasmic reticulum calcium ion transport | aminophospholipid transport            | bile acid and bile salt transport                                                      | response to epinephrine stimulus                                                                 | response to epinephrine stimulus    | regulation of lipid translocation                      | regulation of membrane lipid distribution |
| regulation of chloride transport                                           |                                          |                                                                                                 | microvillus assembly                   | regulation of microvillus organization                                                 | cellular response to epinephrine stimulus                                                        | response to epinephrine stimulus    | regulation of cardiac muscle cell contraction          | xenobiotic transmembrane transport        |
| regulation of chloride transport                                           |                                          |                                                                                                 | regulation of microvillus assembly     | microvillus organization                                                               | cellular response to cAMP                                                                        |                                     |                                                        |                                           |
| dense core granule exocytosis                                              | synaptic vesicle docking                 | dense core granule localization                                                                 |                                        | regulation of cell communication by electrical coupling involved in cardiac conduction | calcium ion-regulated exocytosis of neurotransmitter                                             |                                     | negative regulation of peptidyl-serine phosphorylation |                                           |
| neuronal dense core exocytosis                                             | secretory granule localization           |                                                                                                 |                                        |                                                                                        | calcium ion-regulated exocytosis of neurotransmitter                                             |                                     | establishment of endothelial barrier                   |                                           |
| presynaptic dense core vesicle exocytosis                                  | vesicle docking involved in exocytosis   | vesicle docking                                                                                 |                                        |                                                                                        |                                                                                                  |                                     |                                                        |                                           |

# Vasc-Endo\_0\_downF\_UpM MF



# Vasc-Endo\_0\_upF\_DownM CC

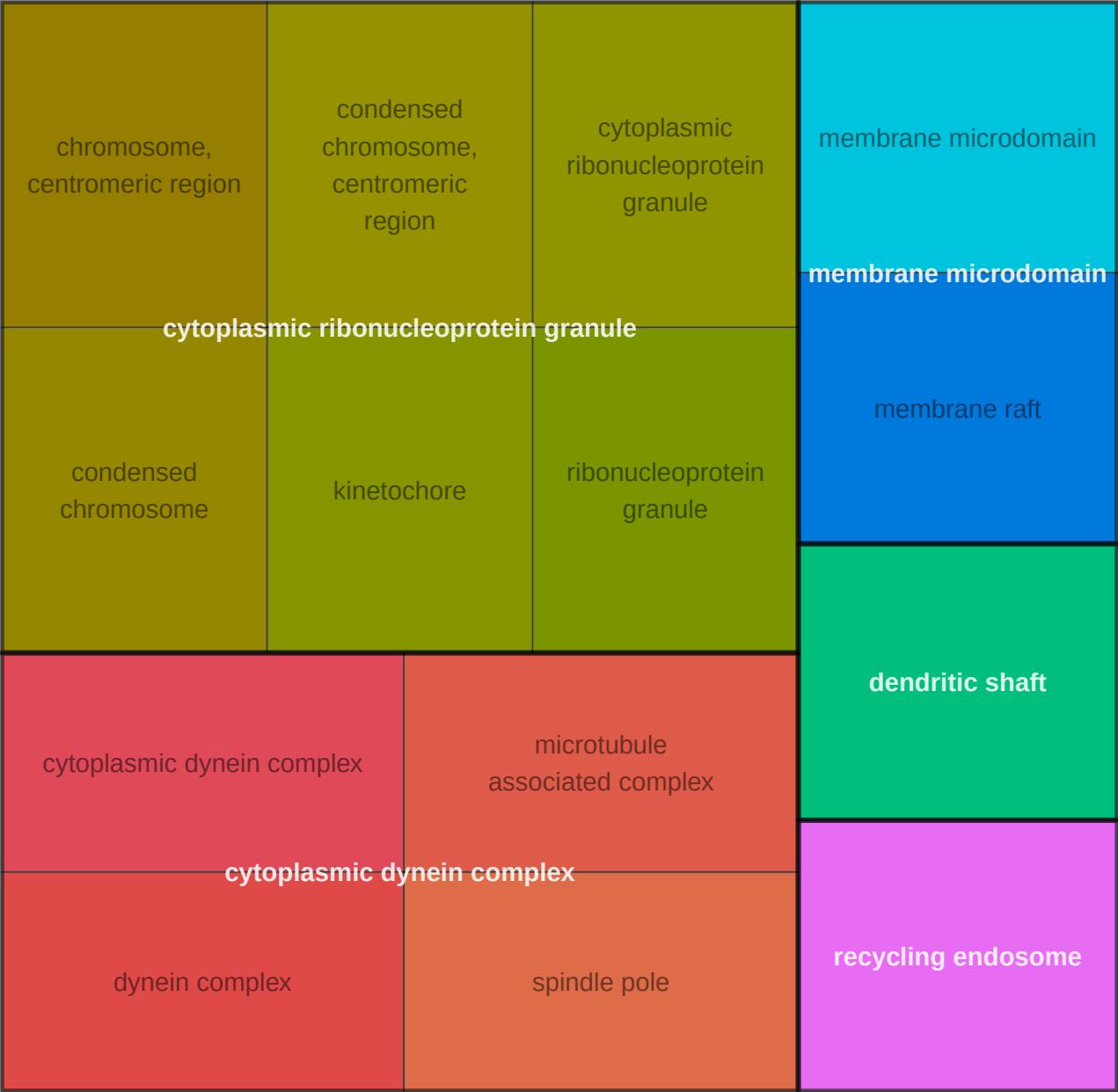


Vasc-Endo\_0\_upF\_DownM BP

|                                                       |                                         |                                               |                                           |                                                |
|-------------------------------------------------------|-----------------------------------------|-----------------------------------------------|-------------------------------------------|------------------------------------------------|
| protein maturation                                    | membrane protein ectodomain proteolysis | sensory perception of light stimulus          | sensory perception of mechanical stimulus | positive regulation of Notch signaling pathway |
| membrane protein proteolysis                          | protein maturation                      | sensory perception of sound                   |                                           | positive regulation of Notch signaling pathway |
| regulation of membrane protein ectodomain proteolysis | protein folding                         | sensory perception of sound                   | visual perception                         | regulation of Notch signaling pathway          |
| chloride transmembrane transport                      | chloride transport                      | peptidyl-histidine modification               |                                           | female pregnancy                               |
| inorganic anion transmembrane transport               | inorganic anion transport               | protein histidyl modification to diphthamide  |                                           | multi-organism reproductive process            |
| inorganic anion transport                             | monoatomic anion transport              | positive regulation of ATP-dependent activity | regulation of ATP-dependent activity      | multi-multicellular organism process           |



Vasc-Endo\_0\_upF\_UpM CC

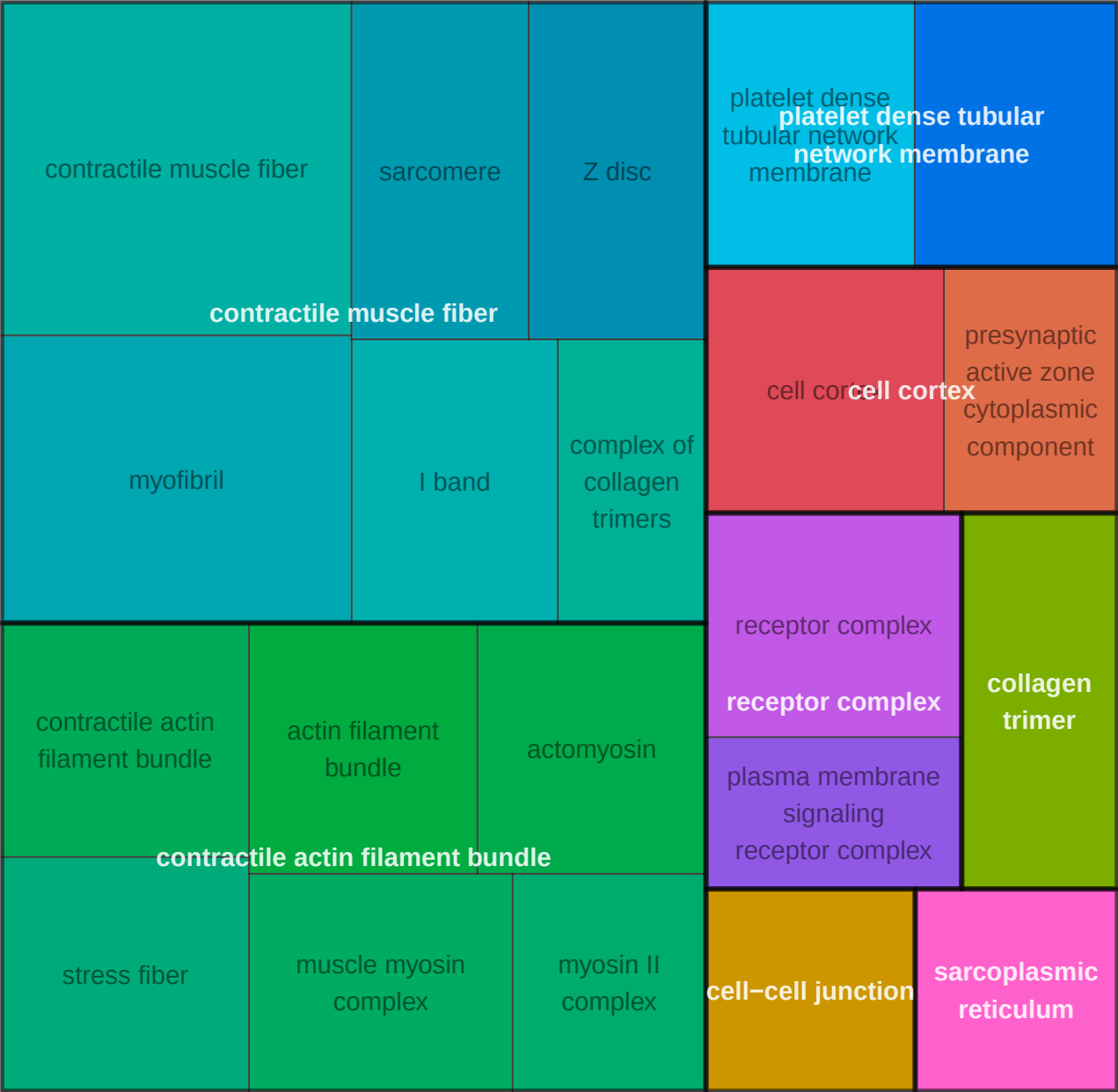


Vasc-Endo\_0\_upF\_UpM BP





Vasc-Endo\_M\_up CC



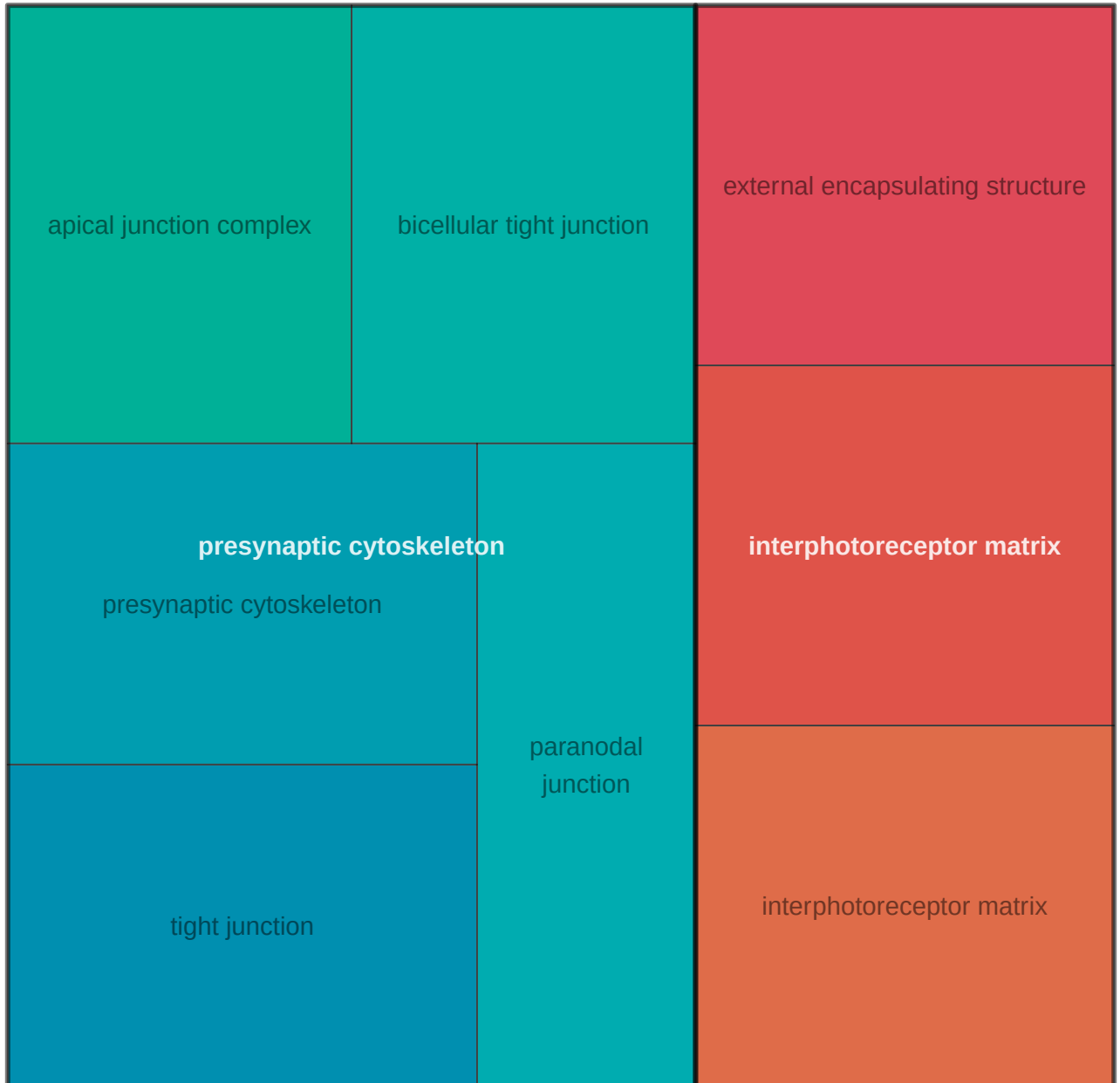
## Vasc-Endo\_M\_up BP



Vasc-Endo\_M\_up MF



# Vasc-VLMC\_0\_downF\_UpM CC



# Vasc-VLMC\_0\_downF\_UpM MF

ciliary neurotrophic  
factor receptor binding

neuroligin family  
protein binding

**calcium-activated  
potassium channel activity**

**ciliary neurotrophic factor receptor binding**

integrin binding

Roundabout binding

**profilin binding**

**laminin-1 binding**

**structural constituent of  
presynaptic active zone**

**GTPase inhibitor activity**

# Vasc-VLMC\_M\_up CC

external encapsulating structure

extracellular matrix

**external encapsulating structure**

**desmosome**

collagen-containing extracellular matrix